Course: COM108 **Quiz 1** Date: 20221007

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1. Evaluate the following limit or show it does not exist. [15%]

(a) $\lim_{x\to 2} \dfrac{\sqrt{x+2}}{x-4}$ (b) $\lim_{x\to 4} \dfrac{2x^3-128}{\sqrt{x}-2}$ (c) $\lim_{x\to 1} \dfrac{\sqrt[3]{x}-1}{\sqrt{x}-1}$

2. Let *f* be the function defined as

$$f(x)=\begin{cases} 2^x & \text{if } x \le 1 \\ 3-x & \text{if } 1 < x \le 4 \\ \sqrt{x} & \text{if } x > 4 \end{cases}$$

(a) Evaluate the limit: $\lim_{x\to 1} f(x)$, if $f(x)$ is continuous at $x=1$. [5%]

(b) Find the value of $a$ and $b$ such that $f(x)$ is continuous on $(-\infty, \infty)$. [5%]

(c) Evaluate the limit: $\lim_{x\to 4} f(x)$. [5%]

3. Find the following limit if it exists. [20%]

(a) $\lim_{x\to\infty} \sqrt{81x^2 + x} - 9x$

(b) $\lim_{x\to-\infty} \dfrac{\sqrt{9x^6 - x}}{x^3 + 1}$